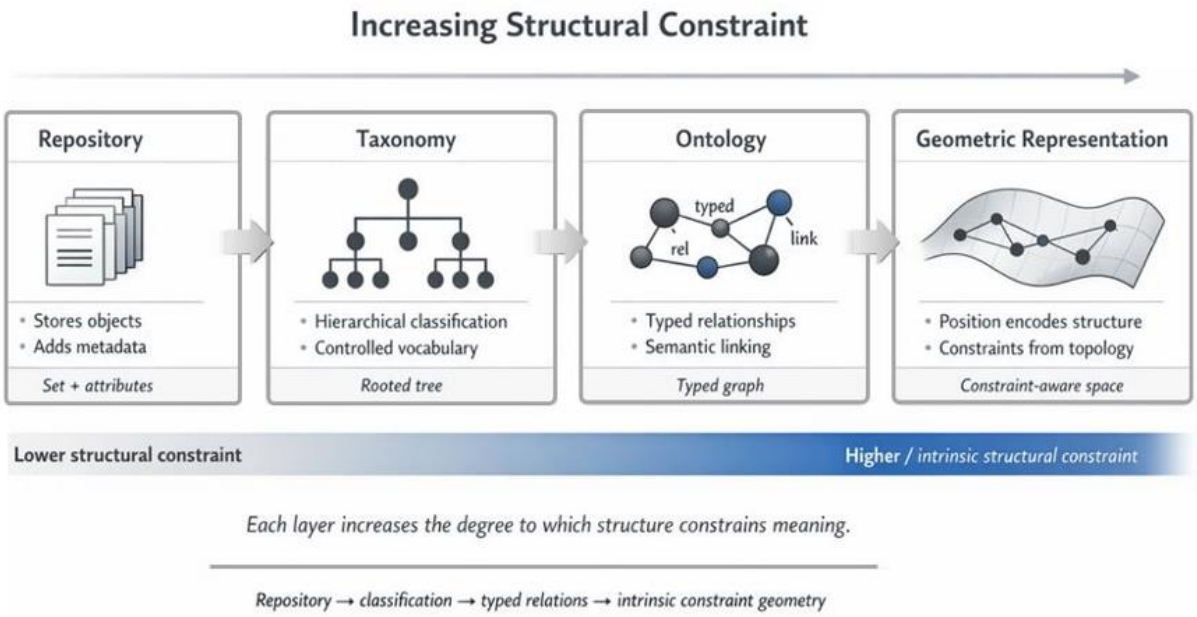

Information storage systems that hold data and metadata but contain minimal inherent structural relationships.	Hierarchical classification that organizes information through controlled vocabularies and category placement.	A semantic model that defines entities and their typed relationships to represent domain knowledge.	The structure through which AI models encode and interpret information within their internal learning space.

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1. Repositories

Foundation Layer: Storage + Metadata

Repositories are the foundational systems where information is stored and retrieved.

They include:

- Databases
- Document stores
- Data lakes and warehouses
- Content management systems
- File systems

Core characteristics:

- Store raw information and records
- Maintain metadata describing documents, datasets, and objects
- Enable indexing and retrieval
- Support access control and governance

At this level, information exists **without formal structural relationships** beyond metadata and schema definitions.

Repositories provide **persistence and access**, but not deep semantic structure.

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2. Taxonomy

Hierarchical Organization & Controlled Vocabulary

Taxonomies introduce structure by organizing concepts into hierarchical categories.

Typical characteristics:

- Parent–child relationships
- Controlled vocabulary terms
- Standardized naming conventions
- Classification trees
- Category placement rules

In enterprise environments, taxonomies enable:

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- Consistent tagging of content and data
- Navigation and discovery
- Harmonization across systems and departments
- Governance of terminology

Taxonomies create **structural constraint through placement** — an item must belong to defined categories within the hierarchy.

They provide the **first explicit structural layer above raw data**.

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3. Ontology

Typed Relationships & Domain Semantics

Ontologies extend taxonomies by introducing **explicit relationships between concepts**.

They typically include:

- Entity types
- Relationship types
- Attributes and properties
- Logical constraints
- Domain rules

Ontologies allow systems to represent knowledge such as:

- Cause and effect relationships
- Functional roles
- Part–whole relationships
- Process dependencies
- Regulatory or domain constraints

In practice, ontologies support:

- Knowledge graphs
- Semantic search
- Data integration across domains
- Advanced reasoning systems

Where taxonomies classify, **ontologies describe how entities relate to one another within a domain**.

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4. Representation

Intrinsic Structural Constraint in AI Models

This layer represents the question explored by AIIA.

Current AI systems typically learn structure **statistically** through embeddings and training data.

However, enterprise systems often require **deterministic constraint** to maintain categorical integrity.

The open question is:

Can structural constraint exist within the representation itself rather than being imposed externally?

Possible characteristics of such systems could include:

- Intrinsic constraint within representation spaces
- Structural invariants encoded in model architecture
- Reduced reliance on external ontology layers for enforcement
- Greater alignment between model reasoning and domain structure

This does **not replace taxonomies or ontologies**, but raises a design question:

Whether future AI systems may carry more structural integrity **inside the model rather than exclusively in external governance layers**.

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